The t_c-Cederbaum model: ghost Gutzwiller by direct minimization

(core fragment with $B_c = 1$, 3 and 5 quasiparticle orbitals)

October 29, 2026

This note defines the Hamiltonian that is solved, the t_c-Cederbaum model (the Cederbaum model of core-hole satellites plus a weak core–virtual hopping t_c; $t_c = 0$ is the original Cederbaum model), its exact solution by diagonalization (ED), the ghost-Gutzwiller (gGut) variational problem used for it, the direct minimization that solves that problem, and the comparison of the gGut results with ED, for a core fragment without ghosts ($B_c = 1$) and with two and four ghosts ($B_c = 3$, 5). The aim of the test is to see whether gGut reproduces the satellite structure of the core-hole spectrum of this model. Quantities are per spin unless stated otherwise; solutions are real and paramagnetic.

1 The Hamiltonian

Three orbitals, c (core), o (occupied) and v (virtual), each with two spins. The t_c-Cederbaum model is $H_{\mathrm{tot}} = H + H_t$, where H is the Cederbaum model (the o–v hopping conditioned on a core hole) and H_t the added hopping:

$$\begin{aligned} H_{\mathrm{tot}} = {} & g \sum_{\sigma} (n_{v\sigma} - n_{o\sigma}) - E_c \sum_{\sigma} n_{c\sigma} - V \sum_{\sigma\sigma'} (1 - n_{c\sigma}) \left(c^{\dagger}_{v\sigma'} c_{o\sigma'} + \mathrm{h.c.} \right) \\ & + t_c \sum_{\sigma} \left(c^{\dagger}_{c\sigma} c_{v\sigma} + \mathrm{h.c.} \right), \end{aligned} \tag{1}$$

with $E_c > 0$, orbital gap g, core–valence coupling V, and a weak core–virtual hopping t_c. The prefactor $\sum_{\sigma}(1 - n_{c\sigma}) = 2 - n_c$ counts the core holes, so the o–v hopping acts only when the core is not full. The particle number and S_z are conserved. The core occupation n_c is conserved only at $t_c = 0$. Energies are measured in units of g ($g = 1$), so the parameters are the ratios E_c/g, V/g and t_c/g. All results below use $E_c/g = 10$, $V/g \in \{1/\sqrt{2}, \sqrt{2}\} = \{0.707, 1.414\}$ and $t_c/g \leq 0.5$; the ground state then lies in the sector with two electrons of each spin. The two values of V are chosen so that $\rho/\Omega = V/(\sqrt{2}\, g) = 0.5$ and 1.0, the ratios of the transition moment ρ to the excitation energy $\Omega = 2g$ used by Kockläuner and Golze for this model [2] (Sec. 6.3); E_c only shifts all poles rigidly at $t_c = 0$.

2 Exact diagonalization

Diagonalization. H_{tot} is diagonalized exactly in the 64-dimensional Fock space of six spin orbitals (Jordan–Wigner operators). At $t_c = 0$ the ground state is the product state $|220\rangle$ (notation $|n_c n_o n_v\rangle$: core and o doubly occupied, v empty) with $E = -2E_c - 2g$ and core occupation

$\nu_c = \langle n_c \rangle = 2$; for $t_c > 0$ it acquires a core-hole component and $\nu_c < 2$. The core-hole Green's function for spin $\uparrow$ is

$$G^h_{cc}(\omega) = \sum_n \frac{|\langle n|c_{c\uparrow}|\mathrm{GS}\rangle|^2}{\omega + (E_n - E_{\mathrm{GS}}) - i\eta}, \tag{2}$$

with poles at $\omega = -E_i$ and weights p_i. At $t_c = 0$ there are three poles; with $\alpha = V/g$,

$$\begin{aligned} E_1 &= E_c + 2g - 2g\sqrt{1+\alpha^2}, & p_1 &= \frac{(1+\sqrt{1+\alpha^2})^2}{4(1+\alpha^2)}, \\ E_2 &= E_c + 2g, & p_2 &= \frac{\alpha^2}{2(1+\alpha^2)}, \\ E_3 &= E_c + 2g + 2g\sqrt{1+\alpha^2}, & p_3 &= \frac{(1-\sqrt{1+\alpha^2})^2}{4(1+\alpha^2)}, \end{aligned} \tag{3}$$

which the ED reproduces to 9×10^{-15}. For $t_c > 0$ the poles move at $O(t_c^2)$ and the middle pole splits into two lines $\sim 10^{-4}$ apart (Tab. 2); Tab. 1 gives the ground-state quantities.

V/g	t_c/g	$(E_0 + 2E_c + 2g)/g$	$2 - \nu_c$	$\rho^{cv} = \langle c^\dagger_{c\uparrow} c_{v\uparrow}\rangle$
0.707	0.5	-4.55×10^{-2}	4.15×10^{-3}	-4.54×10^{-2}
0.707	0.2	-7.30×10^{-3}	6.67×10^{-4}	-1.82×10^{-2}
0.707	0.1	-1.82×10^{-3}	1.67×10^{-4}	-9.12×10^{-3}
0.707	0.05	-4.56×10^{-4}	4.17×10^{-5}	-4.56×10^{-3}
1.414	0.5	-4.60×10^{-2}	4.27×10^{-3}	-4.59×10^{-2}
1.414	0.2	-7.37×10^{-3}	6.87×10^{-4}	-1.84×10^{-2}
1.414	0.1	-1.84×10^{-3}	1.72×10^{-4}	-9.22×10^{-3}
1.414	0.05	-4.61×10^{-4}	4.30×10^{-5}	-4.61×10^{-3}

Table 1: Exact ground state of H_{tot}: $2 - \nu_c \propto t_c^2$ and $\rho^{cv} \propto t_c$.

V/g		$-E_1$	$-E_2$ (split pair)	$-E_3$	p_1
0.707	$t_c \to 0$ (Eq. 3)	−9.5505	−12	−14.4495	0.8249
	$t_c/g = 0.5$	−9.5777	−12.0441, −12.0477	−14.5121	0.8268
	$t_c/g = 0.1$	−9.5516	−12.0018, −12.0019	−14.4520	0.8250
	$t_c/g = 0.05$	−9.5508	−12.0004, −12.0005	−14.4501	0.8249
1.414	$t_c \to 0$ (Eq. 3)	−8.5359	−12	−15.4641	0.6220
	$t_c/g = 0.5$	−8.5707	−12.0406, −12.0510	−15.5198	0.6263
	$t_c/g = 0.1$	−8.5373	−12.0016, −12.0020	−15.4663	0.6222
	$t_c/g = 0.05$	−8.5362	−12.0004, −12.0005	−15.4647	0.6221

Table 2: Exact hole poles of G^h_{cc} at $V/g = 0.707$ and 1.414 (weights $> 10^{-3}$); the split pair has total weight $p_2 = \alpha^2/[2(1+\alpha^2)] = 0.1667$ and 0.3333, respectively.

3 The ghost-Gutzwiller variational problem

Fragments and quasiparticle space. Each orbital $I \in \{c, o, v\}$ is a fragment with B_I quasiparticle orbitals $d_{Ia\sigma}$, $a = 1, \ldots, B_I$: $B_o = B_v = 1$ (no local interaction) and $B_c \in \{1, 3, 5\}$, i.e. $B_c - 1$ ghost orbitals on the core. The Gutzwiller wave function is $|\Psi_G\rangle = \prod_I P_I |\psi_{qp}\rangle$ with $|\psi_{qp}\rangle$

a Slater determinant that fills $N_{\rm occ} = 2 + (B_c - 1)/2$ of the $B_c + 2$ levels per spin. Its one-body density matrix is $\rho^{IJ}_{ab} = \langle d^\dagger_{Ia\sigma} d_{Jb\sigma}\rangle$.

Embedding states. The projector of each fragment is represented by a real normalized state Φ_I in the Fock space of the physical orbital I plus B_I bath orbitals per spin, restricted to the half-filled sector with $(1 + B_I)/2$ electrons of each spin (B_I odd). Writing $c_{I\sigma}$ for the physical and $d_{Ia\sigma}$ for the bath orbitals, the following quantities are expectation values in Φ_I (spin averaged):

$$\begin{aligned} D_{I,a} &= \tfrac{1}{2}\sum_\sigma \langle d^\dagger_{Ia\sigma} c_{I\sigma}\rangle, \quad \Delta^{bb}_{I,ab} = \tfrac{1}{2}\sum_\sigma \langle d^\dagger_{Ia\sigma} d_{Ib\sigma}\rangle, \\ \nu_c &= \langle n_c\rangle, \qquad\qquad\qquad K_{\gamma\delta} = \Big\langle n_c \sum_\sigma d_{c\delta\sigma} d^\dagger_{c\gamma\sigma}\Big\rangle, \end{aligned} \tag{4}$$

and $E^{loc}_I = \langle H^{loc}_I\rangle$ with $H^{loc}_c = -E_c n_c$, $H^{loc}_o = -g n_o$, $H^{loc}_v = +g n_v$ ($n_I = \sum_\sigma n_{I\sigma}$). From them:

$${}^I\Delta = \mathbb{I} - \Delta^{bb}_I, \qquad S_I = \big[{}^I\Delta(\mathbb{I} - {}^I\Delta) + \varepsilon\,\mathbb{I}\big]^{1/2}, \qquad R_I = S_I^{-1} D_I, \qquad C = (\mathbb{I} - {}^c\Delta + \varepsilon\,\mathbb{I})^{-1}, \tag{5}$$

where ε is a small regularizing shift (10^{-9} unless stated) and $R_I \in \mathbb{R}^{B_I}$ is the renormalization vector, $c^\dagger_{I\sigma} \to \sum_a R_{I,a} d^\dagger_{Ia\sigma}$. The non-local core–valence correlation enters through the tensor

$$\bar\Theta = \tfrac{1}{2} S_c^{-1} K S_c^{-1} - \nu_c\, C, \qquad u = \rho^{cv} R_v, \quad w = \rho^{co} R_o, \tag{6}$$

which follows from evaluating $\langle\Psi_G| n_{c\sigma} c^\dagger_{o\sigma'} c_{v\sigma'} |\Psi_G\rangle$ with the Gutzwiller approximation (derived in `main.tex`, Eqs. ThetaEff–ThetaEqn, and `SI.tex`, S2).

Energy functional. The energy is $E = \langle\Psi_G|H_{\rm tot}|\Psi_G\rangle$, evaluated term by term from Eq. (1) with $c^\dagger_{I\sigma} \to \sum_a R_{I,a} d^\dagger_{Ia\sigma}$ and Wick's theorem on $|\psi_{qp}\rangle$.

- *Local terms.* $g(n_v - n_o) - E_c n_c = \sum_I H^{loc}_I$ is evaluated in the embedding states, $\sum_I E^{loc}_I$.
- *Core–virtual hopping.* Per spin, $\langle c^\dagger_{c\sigma} c_{v\sigma}\rangle = R_c^T \rho^{cv} R_v$. The sum over the two spins and the hermitian conjugate give a factor 4: $4t_c\, R_c^T \rho^{cv} R_v$.
- *Conditioned o–v hopping.* The weight $(2 - n_c)$ multiplies $\sum_{\sigma'}(c^\dagger_{v\sigma'} c_{o\sigma'} + \text{h.c.})$ and the expectation value does not factorize, since n_c fluctuates and is correlated with the hopping. Per spin, $\langle (2 - n_c)\, c^\dagger_{v\sigma} c_{o\sigma}\rangle = (2 - \nu_c)\, R_v^T \rho^{vo} R_o + w^T \bar\Theta\, u$: the first part is the mean-field one, with the average number of core holes $2 - \nu_c$; the second is the connected core–valence correlation, built from $\bar\Theta$, u and w defined above. Spin sum and hermitian conjugate again give a factor 4.

Adding the three contributions with the prefactors of Eq. (1):

$$E = \underbrace{\sum_I E^{loc}_I}_{\text{local}} + \underbrace{4t_c\, R_c^T \rho^{cv} R_v}_{t_c\ \text{hopping}} - \underbrace{4V(2 - \nu_c)\, R_v^T \rho^{vo} R_o}_{o\text{–}v\ \text{mean field}} - \underbrace{4V\, w^T \bar\Theta\, u}_{\text{connected}}. \tag{7}$$

Written out with the definitions above, the dependences are: E^{loc}_I, ν_c, Δ^{bb}_I and K are expectation values in Φ_I; $R_I = S_I^{-1} D_I$ with S_I a function of Δ^{bb}_I; $\bar\Theta$ a function of ν_c, K and ${}^c\Delta$; $u = \rho^{cv} R_v$ and $w = \rho^{co} R_o$. So E depends on the embedding states only through E^{loc}_I, D_I, Δ^{bb}_I, ν_c and K, and on $|\psi_{qp}\rangle$ only through ρ.

The limit $t_c = 0$. At $t_c = 0$ the core occupation is conserved and the exact ground state is the product state $|220\rangle$ of Sec. 2, with $\nu_c = 2$. There is no core–valence correlation, so the core is coherent neither with v nor with o: $\rho^{cv} = \rho^{co} = 0$ and $u = w = 0$. Term by term in Eq. (7):

- the t_c hopping vanishes through its explicit factor t_c;
- the mean-field o–v term vanishes through $2 - \nu_c = 0$;
- the connected term vanishes through $u = w = 0$.

Hence

$$E\big|_{t_c=0} = \sum_I E_I^{loc} = -2E_c - 2g, \tag{8}$$

which is the exact energy. Two consequences follow. First, V does not appear: every V term of Eq. (7) is zero. Second, $R_c = S_c^{-1} D_c$ appears in Eq. (7) *only* in the t_c hopping term (the other quantities, E_I^{loc}, ν_c, Δ_I^{bb}, K, do not involve D_c, and u, w contain R_v, R_o but not R_c), so with that term gone E is independent of D_c and the minimization leaves R_c undetermined. At $V = 0$ and $t_c \neq 0$ the situation is different: the two V terms drop out, but $E = \sum_I E_I^{loc} + 4t_c\, R_c^T \rho^{cv} R_v$ still depends on R_c, which is therefore determined by the hopping.

These two consequences are the reason for the hopping t_c. The quantity of interest is not the ground-state energy but the core-hole Green's function G^h_{cc}, whose three poles are final-state physics: the three $N - 1$ eigenstates reached when a core electron is removed and V then moves electrons between o and v. gGut does not compute $N - 1$ states; the spectrum comes from the quasiparticle Hamiltonian H_{qp} (Sec. 4), whose parameters R_I, λ_I are fixed by the ground-state variational problem. At $t_c = 0$ that problem contains no V and leaves R_c free, so the G^h_{cc} built from R_c carries no information about the satellites, although the ground state itself is exact. A weak t_c cures both defects at once: it makes R_c determined and slightly correlates the ground state ($\nu_c < 2$), so that the V terms of Eq. (7) switch on and R_c, H_{qp} acquire V-dependent values. It is a probe, not a change of the question: the exact poles move only at $O(t_c^2)$ (Sec. 2), so a method that captures the satellites should reproduce them at small t_c.

The hopping has to couple $|220\rangle$ to other states, which fixes its form. A hopping $t_c(c_c^\dagger c_o + \text{h.c.})$ is useless: both orbitals are full in $|220\rangle$, so it annihilates the state, and $|220\rangle$ remains an exact eigenstate for every t_c (checked by ED). With $c_v^\dagger c_c$, instead, a core electron moves to the empty v at an energy cost $E_c + g$, so the ground state acquires a core-hole component of amplitude $\sim t_c/(E_c + g)$ and $2 - \nu_c = O(t_c^2)$. This is exactly the combination that Eq. (7) needs to become V-dependent: the term $4t_c\, R_c^T \rho^{cv} R_v$ determines R_c, and $2 - \nu_c \neq 0$ activates the o–v terms.

Variational problem.

$$\min_{\{\Phi_I\},\ \psi_{qp}} E \qquad \text{subject to} \qquad \langle \Phi_I | \Phi_I \rangle = 1, \qquad \rho^{II} = {}^I\Delta(\Phi_I) \quad (I = c, o, v). \tag{9}$$

The second constraint ties the local quasiparticle density matrix to the embedding state; it has $B_I(B_I+1)/2$ independent components per fragment (8 for $B_c = 3$). Multipliers λ_I of this constraint define the quasiparticle Hamiltonian H_{qp} of Sec. 4.

4 Direct minimization

Problem (9) is solved directly (`not_imp_gga_cederbaum.py`), as in the direct-minimization solver used for the Hubbard dimer, without any self-consistency cycle.

Variables. The embedding states are parametrized by unnormalized real vectors x_I in the half-filled sector ($\Phi_I = x_I/|x_I|$; dimensions 36, 4, 4 for $B_c = 3$ and 400, 4, 4 for $B_c = 5$), and the Slater determinant by a real $(B_c + 2) \times N_{\mathrm{occ}}$ matrix X with $\rho = X(X^TX)^{-1}X^T$. The constraints $\rho^{II} = {}^I\Delta(\Phi_I)$ are imposed as equalities (SLSQP, tolerance 10^{-13} on the energy). Everything is evaluated with sector-projected dense matrices of the operators defined in Sec. 3.

Gradients. The analytic gradient of Eq. (7) follows from the chain rule through $R_I(D_I, \Delta_I^{bb})$, $\bar{\Theta}(\nu_c, K, {}^c\Delta)$ and $\rho(X)$:

$$dE = \sum_I 4\, h_I{\cdot}dD_I + \sum_I 2\,\mathrm{Tr}[\Lambda_I\, d\Delta_I^{bb}] + \beta\, d\nu_c + \mathrm{Tr}[\Xi\, dK] + \sum_I dE_I^{loc} + 2\,\mathrm{Tr}[H_{\mathrm{off}}\, d\rho], \tag{10}$$

with $h_I = S_I^{-1}X_I$ and $X_I = \frac{1}{4}\,\partial E/\partial R_I$ given by

$$\begin{aligned} X_c &= t_c\,\rho^{cv}R_v, \qquad X_o = -V\big[(2-\nu_c)\rho^{ov}R_v + \rho^{oc}\bar{\Theta}u\big], \\ X_v &= t_c\,\rho^{vc}R_c - V\big[(2-\nu_c)\rho^{vo}R_o + \rho^{vc}\bar{\Theta}w\big], \end{aligned} \tag{11}$$

The matrices $\Lambda_I = -\frac{1}{2}\nabla^s_{{}^I\Delta}E$ (derivative of E with respect to ${}^I\Delta$ at fixed D_I, through R_I, S_I and $\bar{\Theta}$) are

$$\begin{aligned} \Lambda_{o,v} &= G\big({}^I\Delta,\, h_I R_I^T\big), \\ \Lambda_c &= G\big({}^c\Delta,\, h_c R_c^T - \tfrac{1}{4}Y\big) - V\nu_c\, C\,(uw^T + wu^T)\,C, \\ Y &= V\big(S_c^{-1}K\,\Pi + \Pi\,K\,S_c^{-1}\big), \quad \Pi = xy^T + yx^T, \quad x = S_c^{-1}w,\ y = S_c^{-1}u, \end{aligned} \tag{12}$$

where $G(\Delta, M) = W\,[(\tilde{M} + \tilde{M}^T) \circ F]\,W^T$ is twice the symmetrized gradient of $\mathrm{Tr}[M\,S(\Delta)]$, $\Delta = W\,\mathrm{diag}(n_a)W^T$, $\tilde{M} = W^TMW$, $F_{ab} = (1 - n_a - n_b)/(s_a + s_b)$, $s_a = \sqrt{n_a(1-n_a) + \varepsilon}$ and $\circ$ is the elementwise product. The remaining derivatives are

$$\beta = \frac{\partial E}{\partial \nu_c} = 4V\big(R_v^T\rho^{vo}R_o + w^TCu\big), \qquad \Xi = \nabla_K E = -V\,\Pi, \tag{13}$$

and H_{off} has the off-diagonal blocks $H^{co} = -V(\bar{\Theta}u)R_o^T$, $H^{cv} = t_cR_cR_v^T - V(\bar{\Theta}w)R_v^T$, $H^{ov} = -V(2-\nu_c)R_oR_v^T$ (and their transposes), i.e. $\frac{1}{4}\partial E/\partial\rho^{IJ} = H^{IJ}$. The Jacobian of the constraints is also analytic. Gradients and Jacobian are verified against finite differences in Sec. 5.

Lagrangian and its differential. With the multipliers λ_I (symmetric $B_I \times B_I$, one per independent component of the constraint $\rho^{II} = {}^I\Delta(\Phi_I)$) the Lagrangian of problem (9) is

$$\mathcal{L} = E\big(\{\Phi_I\}, \rho\big) + 2\sum_I \mathrm{Tr}\big[\lambda_I\big({}^I\Delta(\Phi_I) - \rho^{II}\big)\big], \tag{14}$$

where the normalization of Φ_I and the idempotency of ρ carry no multipliers: they are built into the variables, $\Phi_I = x_I/|x_I|$ and $\rho = X(X^TX)^{-1}X^T$. Since ${}^I\Delta = \mathbb{I} - \Delta_I^{bb}$, the constraint term has the differential $-2\mathrm{Tr}[\lambda_I(d\rho^{II} + d\Delta_I^{bb})]$. Adding it to Eq. (10), the terms in $d\Delta_I^{bb}$ combine into $2\mathrm{Tr}[(\Lambda_I - \lambda_I)\,d\Delta_I^{bb}]$ and the terms in $d\rho$ into $2\mathrm{Tr}[(H_{\mathrm{off}} - \mathrm{blockdiag}(\lambda_I))\,d\rho]$:

$$d\mathcal{L} = \underbrace{\sum_I 4\, h_I{\cdot}dD_I + \sum_I 2\,\mathrm{Tr}[\lambda_I^c\, d\Delta_I^{bb}] + \beta\, d\nu_c + \mathrm{Tr}[\Xi\, dK] + \sum_I dE_I^{loc}}_{\text{variation of the } \Phi_I} + \underbrace{2\,\mathrm{Tr}[H_{qp}\, d\rho]}_{\text{variation of } X}, \tag{15}$$

with the two combinations

$$\lambda_I^c = \Lambda_I - \lambda_I, \qquad H_{qp} = H_{\text{off}} - \text{blockdiag}(\lambda_I). \tag{16}$$

These are the bath potential and the quasiparticle Hamiltonian of the self-consistency loop; here they appear as the coefficients of $d\mathcal{L}$. Stationarity of $\mathcal{L}$ is $d\mathcal{L} = 0$ for every variation of the Φ_I and of ρ within the projectors:

- *Variation of X.* $2\text{Tr}[H_{qp}\, d\rho] = 0$ for all variations of the projector, i.e. $[H_{qp}, \rho] = 0$: ρ is built from eigenvectors of its own H_{qp}.
- *Variation of Φ_I.* D_I, Δ_I^{bb}, ν_c, K and E_I^{loc} are expectation values in Φ_I (Sec. 3), so the first group of Eq. (15) is the differential of $\langle\Phi_I|H_I^{imp}|\Phi_I\rangle$ at fixed coefficients, i.e. $2\langle d\Phi_I|H_I^{imp}|\Phi_I\rangle$, with

$$\begin{aligned} H_I^{imp} = H_I^{loc} &+ \sum_a h_{I,a} \sum_\sigma \left(d_{Ia\sigma}^\dagger c_{I\sigma} + c_{I\sigma}^\dagger d_{Ia\sigma}\right) + \sum_{ab} \lambda_{I,ab}^c \sum_\sigma d_{Ia\sigma}^\dagger d_{Ib\sigma} \\ &+ \delta_{Ic}\Big[(\alpha + 2\text{Tr}\Xi)\, n_c - \sum_{\gamma\delta} \Xi_{\gamma\delta}\, n_c \sum_\sigma d_{c\gamma\sigma}^\dagger d_{c\delta\sigma}\Big], \end{aligned} \tag{17}$$

  where $n_c = \sum_\sigma c_{c\sigma}^\dagger c_{c\sigma}$ and $H_c^{loc} = -E_c n_c$, $H_o^{loc} = -g\, n_o$, $H_v^{loc} = +g\, n_v$. The coefficients are

$$h_I = S_I^{-1} X_I, \qquad \lambda_I^c = \Lambda_I - \lambda_I, \qquad \alpha = \beta = 4V\big(R_v^T \rho^{vo} R_o + w^T C u\big), \qquad \Xi = -V\,\Pi, \tag{18}$$

  with X_I, Λ_I, Π, β as in Eq. (10) and below. Each term is the coefficient of one expectation value in the first group of Eq. (15): $h_{I,a}$ multiplies $\langle\sum_\sigma (d_{Ia\sigma}^\dagger c_{I\sigma} + \text{h.c.})\rangle = 4D_{I,a}$; $\lambda_{I,ab}^c$ multiplies $\langle\sum_\sigma d_{Ia\sigma}^\dagger d_{Ib\sigma}\rangle = 2\Delta_{I,ab}^{bb}$; $\alpha + 2\text{Tr}\Xi$ multiplies ν_c; and $-\Xi_{\gamma\delta}$ multiplies $\langle n_c \sum_\sigma d_{c\gamma\sigma}^\dagger d_{c\delta\sigma}\rangle = 2\nu_c\delta_{\gamma\delta} - K_{\gamma\delta}$, which with the $2\text{Tr}\Xi\,\nu_c$ part reproduces $\text{Tr}[\Xi\, dK]$. A constant from normal ordering, $-2\text{Tr}\lambda_I^c$, is dropped. Its vanishing for all $d\Phi_I \perp \Phi_I$ means that Φ_I is an eigenstate of H_I^{imp}. It is not the lowest one: for $B_c = 3$, $t_c/g = 0.5$ the direct-minimization Φ_c is the fourth ($V/g = 0.707$) and the sixth ($V/g = 1.414$) of the 36 eigenstates of H_c^{imp} in energy order, with the lowest eigenvalue 2.3 and 8.1 below it, whereas Φ_o and Φ_v are ground states (`excited_phi_cederbaum.tex`, `excited_check_golze.py`). The impurity loop, which takes the lowest state, cannot follow it.

The self-consistency loop enforces these conditions one after the other (diagonalizing H_{qp}, solving for Φ_I); the direct minimization does not diagonalize either. It passes E, dE (Eq. (10)) and the constraints $\kappa = \rho^{II} - {}^I\Delta(\Phi_I)$ with their Jacobian ($d\kappa = d\rho^{II} + d\Delta_I^{bb}$) to SLSQP, which drives the gradient to a constrained stationary point. The multipliers are recovered afterwards from the first group of Eq. (15): stationarity in Φ_I of $E + \sum_k \mu_k\kappa_k$ is solved for μ in the least-squares sense (below). The eigenstate Φ_I is not required to be the lowest one of H_I^{imp}.

Note. In standard gGA, H_I^{imp} contains only H_I^{loc}, the hybridization h_I and the bath term λ_I^c, and the other fragments enter through h_I alone (the t_c hopping does this). The conditioned o–v hopping multiplies a bilinear by n_c, so its average also depends on ν_c and K (through $\bar{\Theta}$, from `main.tex`, `SI.tex`). This adds the n_c and $n_c d^\dagger d$ terms of H_c^{imp} and makes its coefficients α, Ξ, λ_c^c depend on o and v (through u, w, ρ^{vo}), and h_o, h_v on the core.

Starting points. The problem is non-convex, so several random starts are run per parameter set (embedding states from random small Hamiltonians, X from a random symmetric matrix); a point is accepted if the constraint residual is $< 10^{-9}$, and the lowest accepted energy is kept. Seeding from the previous t_c can end in a poor local minimum (Sec. 6.1), so points are judged by their residuals and not by the energy alone.

$B_c = 5$**.** Random starts rarely converge at $B_c = 5$ (436 variables), so the search is seeded from a $B_c = 3$ solution with the two new ghosts asleep: $\Phi_c^{(5)} = \Phi_c^{(3)} \otimes |\text{ghost}\rangle$ (one extra bath orbital filled, one empty) and $\rho^{(5)} = \rho^{(3)} \oplus \text{diag}(0, 1)$. This point is feasible and has the $B_c = 3$ energy; it is perturbed randomly and minimized. Sparse operators and caching make a $B_c = 5$ run take about 2 s.

Quasiparticle Hamiltonian and spectrum. The multipliers are recovered from the stationarity of $E + \sum_k \mu_k \kappa_k$ in Φ (least squares over the constraint components $\kappa_k = \rho^{II}_{ab} - {}^I\Delta_{ab}$; the residual of this fit is reported as the KKT residual), with $\lambda_{aa} = -\mu_{aa}/2$, $\lambda_{a<b} = -\mu_{ab}/4$. The quasiparticle Hamiltonian is $H_{qp} = H_{\text{off}} - \text{blockdiag}(\lambda_I)$, whose commutator with ρ vanishes at a solution (reported as $\|[H_{qp}, \rho]\|$). The hole Green's function of the core orbital is

$$G^h_{cc}(\omega) = \sum_{n \le N_{\text{occ}}} \frac{|R_c^T \phi_n^{(c)}|^2}{\omega - \epsilon_n - i\eta}, \tag{19}$$

with ϵ_n, ϕ_n the eigenvalues (ascending) and eigenvectors of H_{qp} and $\phi_n^{(c)}$ the core block of ϕ_n; the poles ϵ_n are compared with $-E_i$ and the weights with p_i [Eq. (3)]. The total weight of G_{cc} is $R_c^T R_c$. The constraint Jacobian is rank deficient at solutions with pure ghost occupations (eigenvalues 0 or 1 of ${}^c\Delta$): some multiplier components are then undetermined (Sec. 6.2).

5 Checks

The analytic gradients of Sec. 4 agree with central finite differences of Eq. (7) at random points, for $B_c = 1, 3, 5$, to $\le 4 \times 10^{-10}$ for each identity (H^{IJ}, $S_I h_I$, $G(\Delta, M)$, the ${}^c\Delta$ gradient at fixed R_c, $\beta = \partial E/\partial \nu_c$, $\Xi = \nabla_K E$), and the complete gradient of the minimization to a relative error of 4×10^{-7} (limited by the stiffness of S^{-1} for nearly pure ${}^I\Delta$); the constraint Jacobian to 3×10^{-10}. The limits are reproduced: at $V = 0$, $E - E_{\text{ref}} = +4.4 \times 10^{-8}$ for $B_c = 1$ and 3 (the effect of $\varepsilon = 10^{-9}$; $\nu_c = 1.995893$ and 1.999835 at $t_c/g = 0.5$ and 0.1); at $t_c = 0$ the energy is exact ($< 10^{-15}$ at $B_c = 1$; 10^{-13} and 10^{-11} at $B_c = 3$ for $V/g = 0.707$ and 1.414) and $\nu_c = 2$. At $t_c = 0$, R_c is not determined by the stationarity conditions and V does not enter (Sec. 3). The accepted $B_c = 3$ and 5 points have constraint residual $\lesssim 10^{-9}$ (typically 10^{-14}) and KKT residual $\le 2.5 \times 10^{-6}$.

6 Results for $B_c = 1$, 3 and 5

$B_c = 3$ and 5 mean two and four ghost orbitals on the core, $B_c = 1$ none (standard Gutzwiller). $B_c = 3$ and 5 are from the direct minimization ($B_c = 5$ seeded from the $B_c = 3$ solution, plus random starts); $B_c = 1$ from the converged loop, which the direct minimization reproduces to $\le 1.5 \times 10^{-14}$ in E at $t_c/g = 0.1$. All solutions used in the figures and tables are saved (`cederbaum_data/ghost_solutions.npz`).

6.1 Energy and ν_c

- $B_c = 1$**:** $E_{\text{gGut}} - E_{\text{ED}} \simeq 1.3 \times 10^{-3}\, V^2 t_c^2$ over $t_c/g \in [0.025, 0.5]$ (exponent 1.97 and 1.99), and $|\nu_c - \nu_c^{\text{ED}}| \propto t_c^2$.
- $B_c = 3$**:** $E_{\text{gGut}} - E_{\text{ED}} = 9.2$–$9.6 \times 10^{-8}$ for all (V, t_c), flat; $|\nu_c - \nu_c^{\text{ED}}| \le 2.3 \times 10^{-8}$. This floor is the regularization shift, not the ansatz.

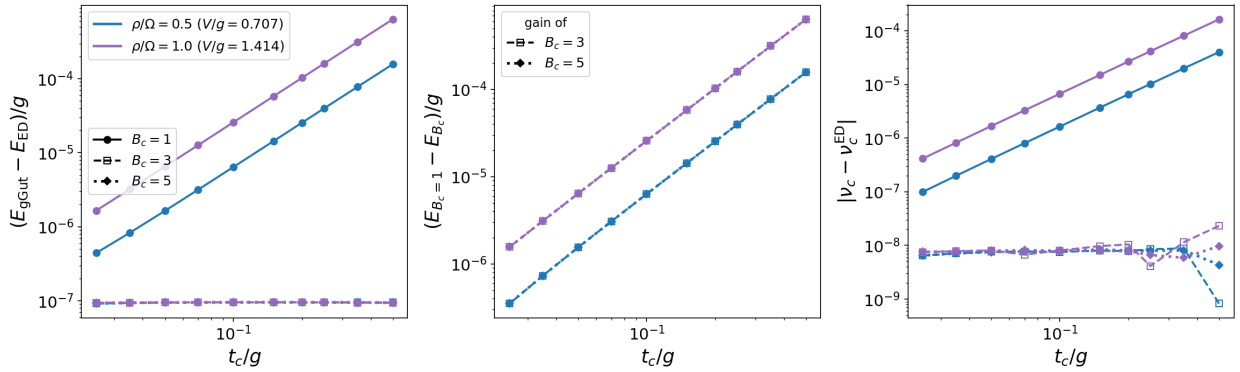


Figure 1: Left: $E_{\mathrm{gGut}} - E_{\mathrm{ED}}$ for $B_c = 1$ (full lines), $B_c = 3$ (dashed, open squares) and $B_c = 5$ (dotted, diamonds); blue: $\rho/\Omega = 0.5$ ($V/g = 0.707$), purple: $\rho/\Omega = 1.0$ ($V/g = 1.414$). Middle: energy gained by the ghosts, $E_{B_c=1} - E_{B_c}$ (the two coincide). Right: $|\nu_c - \nu_c^{\mathrm{ED}}|$. Log–log axes; $E_c/g = 10$.

- $B_c = 5$**:** the same energy at all 20 points ($E_{B_c=5} - E_{B_c=3}$ between -6×10^{-11} and $+6 \times 10^{-13}$; KKT $\leq 2.3 \times 10^{-6}$); 5–9 of 14 runs per point converged to the same energy, and 3 of 120 random starts. The extra ghosts do not lower the energy.

The ghosts thus recover 79–99.99% of the $B_c = 1$ error (the smallest fractions at the smallest t_c, where the shift floor is comparable to the error). Poor local minima from continuation ($(V/g, t_c/g) = (0.707, 0.025)$, $(1.414, 0.07)$) were recomputed from random starts; one $B_c = 1$ loop point needed mixing 0.3; two $B_c = 5$ seeds were obtained by continuation.

6.2 Hole spectral function

		ED main pole		gGut $B_c = 1$		gGut $B_c = 3$		gGut $B_c = 5$	
V/g	t_c/g	position	weight	position	weight	position	weight	position	weight
0.707	0.1	−9.5516	0.8250	−10.0007	0.9999	−9.9355	0.9940	−9.9355	0.9940
0.707	0.05	−9.5508	0.8249	−9.9997	0.9999	−9.9330	0.9939	−9.9330	0.9939
1.414	0.1	−8.5373	0.6222	−10.0005	0.9999	−9.7458	0.9767	−9.7458	0.9767
1.414	0.05	−8.5362	0.6221	−9.9997	0.9999	−9.7435	0.9766	−9.7434	0.9766

Table 3: Dominant hole pole of G_{cc}. ED additionally has satellites at about −12.0 (total weight 0.167 and 0.333 for $V/g = 0.707$, 1.414) and at −14.45 and −15.47 (weights 0.008 and 0.045); gGut has none: all other occupied levels have weight $\leq 1.5 \times 10^{-6}$ ($B_c = 1$) and $\leq 2.2 \times 10^{-6}$ ($B_c = 3$, 5).

Result. The $B_c = 1$ pole sits at −10.000 for every V: the bare core level. At $B_c = 3$ and 5 (identical to 10^{-4}) the single pole moves with V towards the exact main pole but covers only 15–18% of the exact displacement from $-E_c$ (0.067 of 0.449 and 0.257 of 1.464 at $t_c/g = 0.05$), and its weight (0.994, 0.977) stays far above the exact one (0.825, 0.622). *No satellites appear*; the ghosts only add V dependence to the main pole.

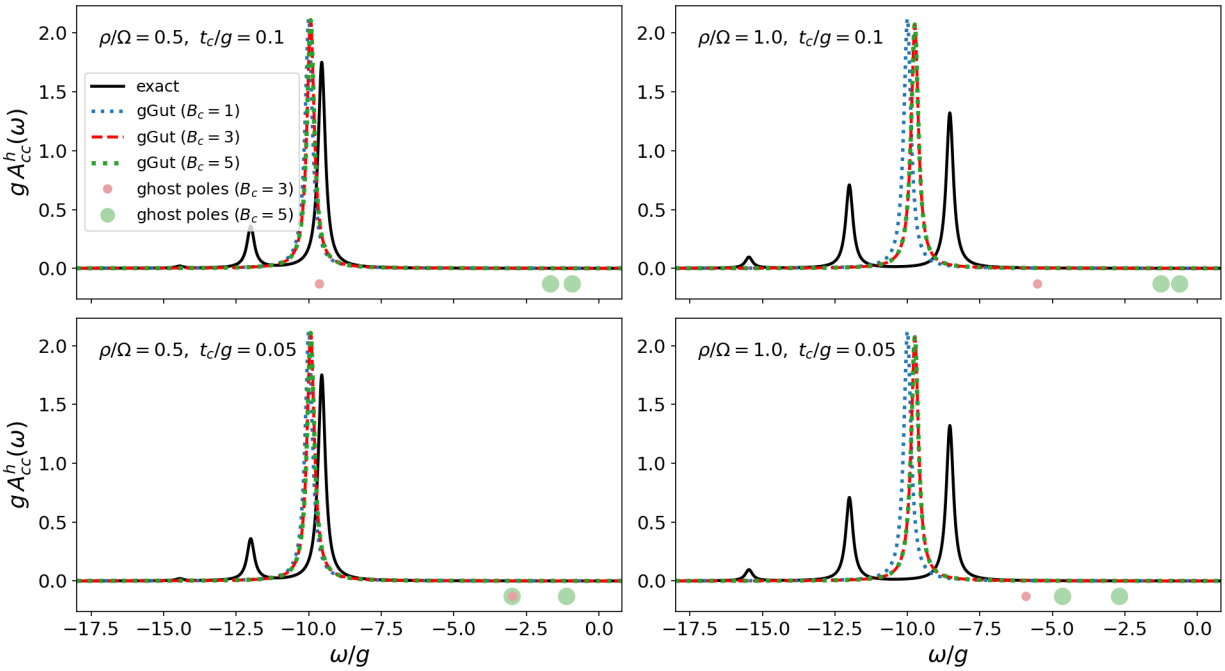


Figure 2: Hole spectral function $A^h_{cc}(\omega) = \sum_n w_n\,(\eta/\pi)/[(\omega-\omega_n)^2+\eta^2]$, $\eta = 0.15$: exact (solid), gGut $B_c = 1$ (blue dotted), $B_c = 3$ (red dashed), $B_c = 5$ (green dotted, on top of $B_c = 3$). Rows: $t_c/g = 0.1$, 0.05; columns: $\rho/\Omega = 0.5$ ($V/g = 0.707$), 1.0 ($V/g = 1.414$); $E_c/g = 10$. Dots below zero: the occupied muted poles (zero-weight ghost levels of H_{qp}, defined in the text), as returned by the minimizer: one at $B_c = 3$ (small, rose), two at $B_c = 5$ (large, green); $B_c = 1$ has none. Their positions are not fixed by the solution (see text) and they carry no weight.

Why there is no weight on the ghosts (interpretation). The weights in Eq. (19) are $|R_c^T \phi_n^{(c)}|^2$. At $B_c = 3$ ($t_c/g = 0.1$) the natural orbitals of ${}^c\Delta$ have occupations $\sim 10^{-7}$, 0.9999 and 1 (the nearly full core orbital and two pure ghosts), and R_c lies entirely along the core orbital (components along the others $\leq 10^{-4}$), so the occupied ghost level carries 3×10^{-7} and 4×10^{-10} at $V/g = 0.707$, 1.414 ($\leq 2.3\times10^{-9}$ at $B_c = 5$), and $R_c^T R_c = 0.994$, 0.977 sits on one pole. Mixing the two filled core orbitals by hand in λ_c does split the weight (e.g. 0.57/0.42 at $V/g = 0.707$, $\delta = 1$), but $\|[H_{qp},\rho]\|$ then grows from $\leq 2.6\times10^{-5}$ to 10^{-2}–4×10^{-2}: these points are not stationary, and their poles and weights do not match the exact ones. The ghosts appear to carry the correlation through ν_c and K (the terms $-V(\bar\Theta u)R_o^T$, $-V(\bar\Theta w)R_v^T$ of H_{qp}) but not through R_c, which is all that Eq. (19) sees.

Muted poles. In gGut every pole of the spectral function is an eigenvalue of H_{qp}, which has $B_c + B_o + B_v$ levels. Three of them are the quasiparticle (QP) levels of c, o and v, the levels where the orbital c, o or v has its largest weight: the core hole near -10, the occupied level near $-g$ and the virtual level near $+g$. The remaining $B_c + B_o + B_v - 3$ levels are the *muted poles*: two at $B_c = 3$ (with $B_o = B_v = 1$) and four at $B_c = 5$. Occupied ones are hole poles, empty ones particle poles. Their weight $|R_I^T \phi_n^{(I)}|^2$ is $\leq 2.2\times10^{-6}$ on every orbital, so they are invisible in $A(\omega)$. They matter because they are the only place where gGut could put a satellite: a satellite needs a pole of H_{qp} at $\omega \simeq -12$ or lower, with weight.

- *Character.* The hole-side ones are ghost levels of the core block (> 0.99), with core weight $< 10^{-4}$. They are counted by occupation, not by energy order, since a free ghost level can cross the level o (in the $B_c = 5$ solution at $t_c/g = 0.05$, $V/g = 1.414$ the o level is empty and a ghost level is occupied). The Fig. 2 dots are the occupied ones: one at $B_c = 3$, two at $B_c = 5$. The empty level at $\omega \simeq +12$ (Fig. 3) is not a ghost: its weight sits on o ($\simeq 10^{-7}$–10^{-6}) and it matches the exact o particle peak at 12.003.

- *Position.* It is not fixed by the solution: the constraint Jacobian has rank 7 of 8 ($B_c = 3$) and 13 of 17 ($B_c = 5$), and shifting the multipliers along the undetermined directions moves the ghost levels, for example to the exact satellite energies, without changing E, the main pole, its weight or $\|[H_{qp}, \rho]\|$; their weights stay $\leq 8 \times 10^{-9}$. Different minimizer solutions of the same point place them differently (-2.98 and -5.93 in the figures, -2.20 and -9.53 in earlier solutions, at $B_c = 3$, $t_c/g = 0.05$); the returned values are the minimum-norm multipliers. They lie between the core level (≈ -10) and the valence region (≈ -1), because the ghost levels come from the multipliers of the density constraint, which are of valence-scale magnitude (the bath levels of λ^c_c at $t_c/g = 0.5$ are -2.1 to $+1.7$ and -5.1 to $+3.0$; `excited_phi_cederbaum.tex`), and not from E_c.

- *Weight.* It is set by R_c, which the energy minimization fixes along the core orbital. A solution with weight on a muted pole would have a different R_c and so differ from the minimum; none was found.

- *Ghosts on o and v.* With $B_c = B_o = B_v = 3$ (9 levels, 6 muted poles) I seeded the minimization from the $B_c = 3$ solution plus a sleeping ghost pair on o and on v (same energy), perturbed it (15 runs per case), and also tried random starts. No run went below the seed energy; the random starts mostly fell to the trivial product state ($R_c = 0$). In three of the four cases of Fig. 3 a run returns to the $B_c = 3$ point ($E - E_{\mathrm{ED}} \simeq 9.5 \times 10^{-8}$, same main pole and weight); at $V/g = 0.707$, $t_c/g = 0.1$ the best point found is $B_c = 1$-like (6.4×10^{-6}). The extra poles again carry no weight (purple row of Fig. 3). The search is limited (SLSQP, seeds from $B_c = 3$ only).

The exact satellites lie *below* the main peak (-12 and -14.5 at $V/g = 0.707$, -15.5 at $V/g = 1.414$, against -9.6 exact, -9.9 gGut), whereas the muted poles lie above it, so none of them can play the role of a satellite.

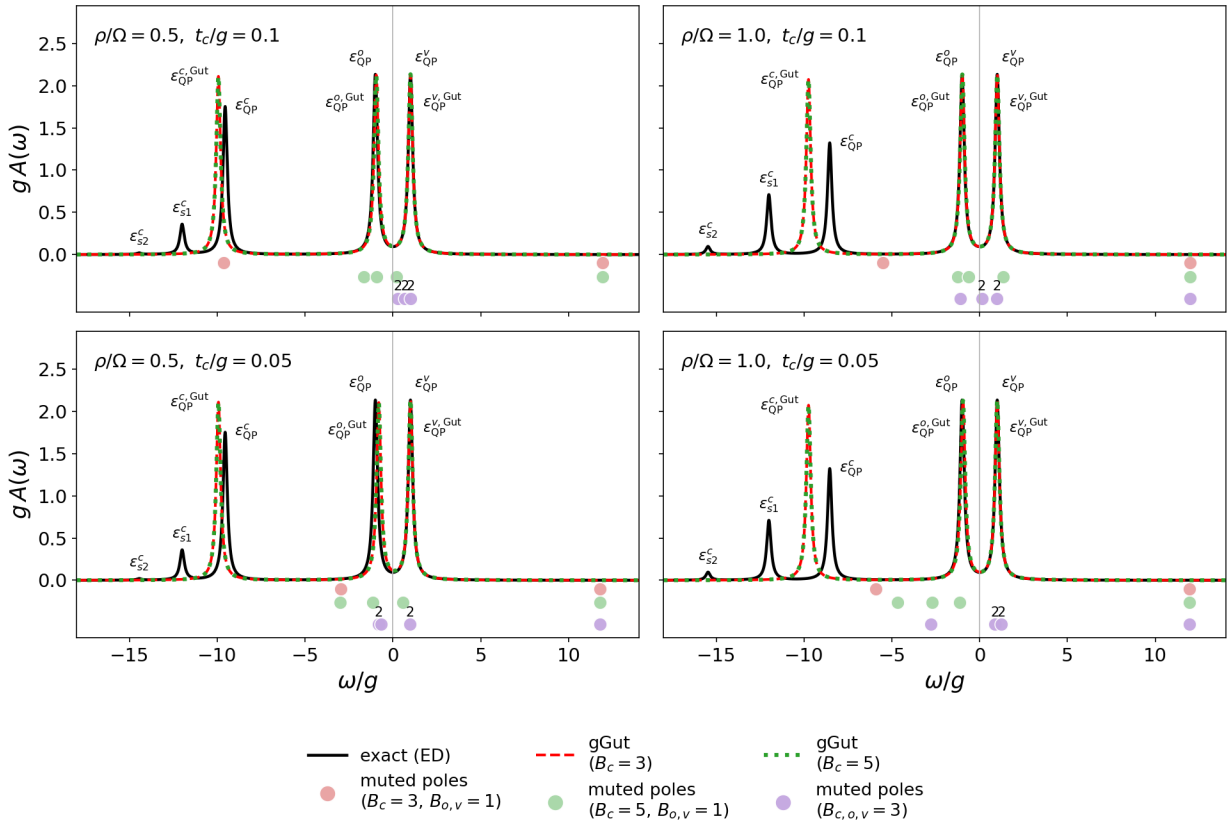


Figure 3: Full spectral function $A(\omega) = \sum_{I=c,o,v} A_{II}(\omega)$ (hole side $\omega < 0$, particle side $\omega > 0$; per spin, $\eta = 0.15$, $E_c/g = 10$). Rows: $t_c/g = 0.1$, 0.05; columns: $\rho/\Omega = 0.5$, 1.0. *Curves:* exact (black), gGut $B_c = 3$ (red dashed) and $B_c = 5$ (green dotted; both with $B_o = B_v = 1$, from the cached solutions of Fig. 2). *Labels:* $\varepsilon^I_{\mathrm{QP}}$ is the quasiparticle peak of the exact spectrum for the orbital $I = c, o, v$ and ε^c_{s1}, ε^c_{s2} are the exact core satellites, numbered by increasing distance from $\varepsilon^c_{\mathrm{QP}}$; $\varepsilon^{I,\mathrm{Gut}}_{\mathrm{QP}}$ are the gGut quasiparticle peaks (the core one is single, with no satellites). *Dots below the axis:* the muted poles of gGut, i.e. all eigenvalues of H_{qp} except the three quasiparticle levels. One row per solution: $B_c = 3$ (rose, 2 poles), $B_c = 5$ (green, 4) and $B_c = B_o = B_v = 3$ (purple, 6; the lowest-energy run seeded from the $B_c = 3$ solution with sleeping ghosts on o and v; in the panel $\rho/\Omega = 0.5$, $t_c/g = 0.1$ no run returned to the seed and the best point is $B_c = 1$-like). Poles closer than 0.02 share one dot, with the multiplicity written above it when it is larger than 1. The muted poles carry no weight ($\leq 2.2 \times 10^{-6}$) and none lies near the exact satellites.

Full spectral function (core, occupied, virtual). Fig. 3 shows all peaks of $A = A_{cc} + A_{oo} + A_{vv}$, hole side ($\omega < 0$) and particle side ($\omega > 0$), against ED. In gGut every eigenvalue of H_{qp} is a pole, occupied ones on the hole side and empty ones on the particle side, with weight $|R_I^T \phi_n^{(I)}|^2$ on orbital I. The three main peaks are the core hole near -10 (c), the occupied level near $-g$ (o) and the virtual level near $+g$ (v); gGut reproduces the last two, and the core pole only approximately (position and weight as in Tab. 3). What gGut misses are the exact core satellites at -12 and -14.5 (-15.5); the remaining ED peaks ($\omega \simeq 3$ for c and $\omega \simeq 12$ for o, particle side) have weight $\lesssim 10^{-6}$ and are not visible. The remaining levels are the muted poles defined above ($B_c = 3$: one hole pole at -5.9 to -3.0 and the empty level at $\simeq +12$; $B_c = 5$: several between -4.7 and $+1.4$ besides that one), all with weight $\leq 10^{-6}$.

6.3 Comparison with the three-level model of Cederbaum and of Kockläuner and Golze

The orbitals c, o, v are the three-level core-hole model of Cederbaum, Domcke and Schirmer [1] (Eqs. (28)–(29)), used by Kockläuner and Golze [2] (Fig. 2) to test G_0W_0 and GW plus cumulant (GW+C). With $\Omega = \epsilon_v - \epsilon_o = 2g$ and the transition moment ρ (GW+C excitation weight $\gamma = \rho^2/\Omega^2$), the exact solution is the 3×3 matrix with off-diagonal elements $-\sqrt{2}V$, which gives Eq. (3) and the weights $(a \pm \Omega)^2/4a^2$, $2V^2/a^2$ of Ref. [1] ($a = (\Omega^2 + 4V^2)^{1/2}$). The $\sqrt{2}$ is the normalization of the two spin channels, so

$$\rho = \sqrt{2}\, V, \qquad \frac{\rho}{\Omega} = \frac{V}{\sqrt{2}\, g}, \tag{20}$$

confirmed by the small-V limit of the exact satellite-to-QP weight ratio ($2.00\times 10^{-4} = (\sqrt{2}V)^2/\Omega^2$ at $V/g = 0.02$, `g0w0_cederbaum.py`). Their ratios $\rho/\Omega = 0.5$, 1.0 are $V/g = 0.707$, 1.414, the values used throughout this note. Their figure gives only the ratio, no energy axis, so E_c cannot be taken from it (it only shifts the poles rigidly); Fig. 4 therefore uses ω with ticks at the exact ϵ_c^{QP}, $\epsilon_c^{\mathrm{Sat1}}$, $\epsilon_c^{\mathrm{Sat2}}$.

G_0W_0 and GW+C for the model (computed here). We computed G_0W_0 and GW+C for this model ourselves (`g0w0_cederbaum.py`) from the expressions below, in order to compare with the G_0W_0 and GW+C of Ref. [2] (not taken from it). With G_0 the bare core level and one bosonic mode, $\Sigma_c(\omega) = \rho^2/(\omega - \epsilon_c + \Omega)$ (no exchange or Hedin shift), the poles of $[\omega - \epsilon_c - \Sigma_c]^{-1}$ solve $x(x+\Omega) = \rho^2$: a QP pole and one satellite at $-\Omega\sqrt{1 + 4\rho^2/\Omega^2}$, against the exact satellites at $-\Omega\sqrt{1 + 2\rho^2/\Omega^2}$ and twice that. GW+C has Poisson satellites at $\epsilon_{\mathrm{QP}} - n\Omega$ with weights $e^{-\gamma}\gamma^n/n!$. These expressions agree with Eqs. (11), (14), (16)–(17), (33), (36), (44)–(47) of Ref. [2] (the exchange constant $\Sigma^x - v^{xc}$ is absorbed in ϵ_c), and the first-satellite positions measured in their rendered Fig. 2b,c (in units of Ω, accuracy a few per cent) agree with this note: exact 1.22 and 1.76 (here 1.225, 1.732), G_0W_0 1.42 and 2.23 (here 1.414, 2.236) at $\rho/\Omega = 0.5$, 1.0.

ρ/Ω		satellite positions /Ω (from QP)	weights (QP, satellites)
0.5	exact	-1.225, -2.449	0.825, 0.167, 0.008
	G_0W_0	-1.414	0.854, 0.146
	GW+C	-1, -2, -3, …	0.779, 0.195, 0.024, …
1.0	exact	-1.732, -3.464	0.622, 0.333, 0.045
	G_0W_0	-2.236	0.724, 0.276
	GW+C	-1, -2, -3, …	0.368, 0.368, 0.184, …

Table 4: Exact ($t_c = 0$), G_0W_0 and GW+C hole spectrum of the model at the ratios of Fig. 2 of Ref. [2] ($V/g = 0.707$, 1.414).

What matches. The exact spectrum has the QP peak and two satellites (a single and a double $o \to v$ excitation, the second much weaker), with intensity and QP–satellite separation growing with ρ/Ω; G_0W_0 has one satellite at higher binding energy than the exact one ($-1.41\,\Omega$ against $-1.23\,\Omega$ at $\rho/\Omega = 0.5$; $-2.24\,\Omega$ against $-1.73\,\Omega$ at 1.0); GW+C puts satellites at $-n\Omega$ (closer to the exact ones at $\rho/\Omega = 0.5$ than at 1.0) and shows spurious higher satellites at larger ρ/Ω; all as stated in Ref. [2]. The hopping t_c moves the exact poles by at most 0.06 at $t_c/g = 0.5$.

What does not match. gGut gives a single pole at both ratios (weight 0.977–0.9999): it reproduces neither the exact satellites nor the GW+C or G_0W_0 ones, and does worse than G_0W_0 for

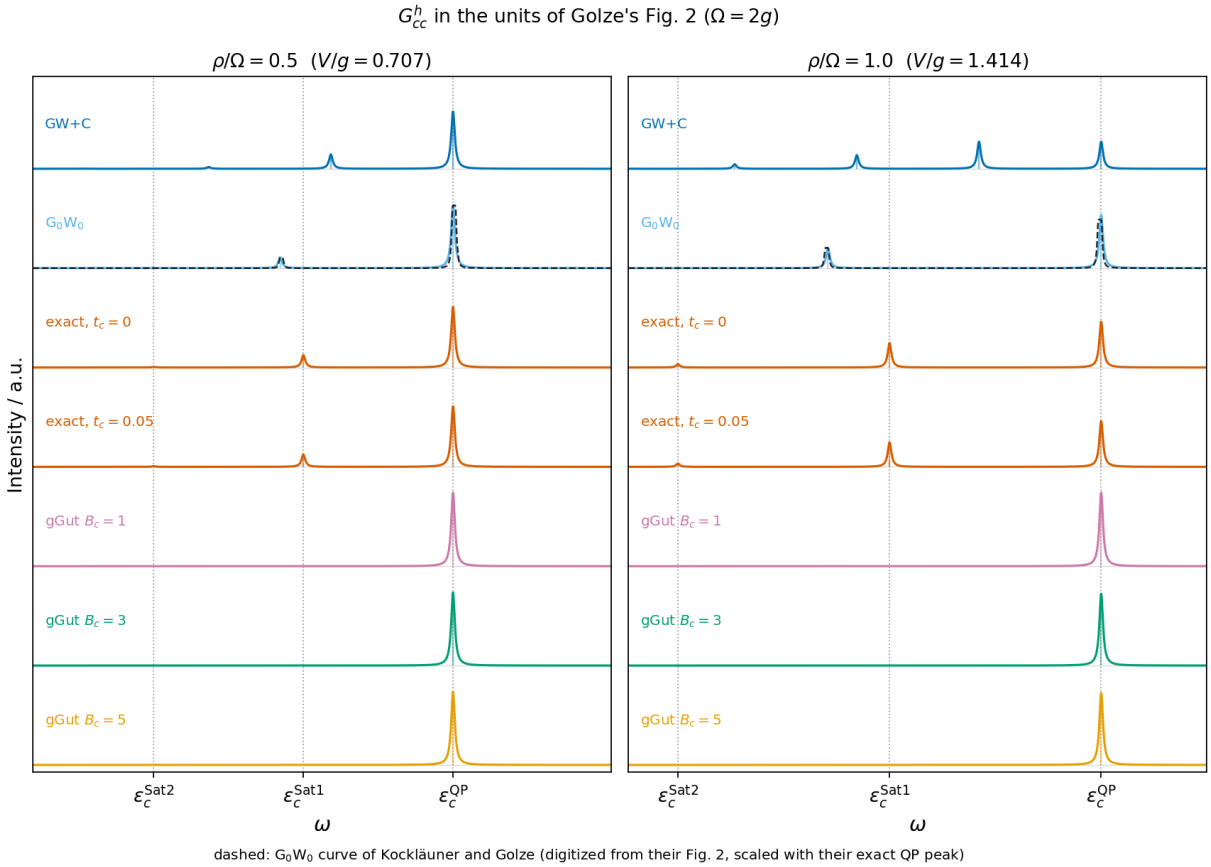


Figure 4: G^h_{cc} in the units of Fig. 2 of Ref. [2]; left $\rho/\Omega = 0.5$ ($V/g = 0.707$), right $\rho/\Omega = 1.0$ ($V/g = 1.414$), $E_c/g = 10$. Rows: GW+C; G_0W_0; exact at $t_c = 0$ and 0.05; gGut $B_c = 1$, 3, 5 ($t_c/g = 0.05$). All rows share one scale (peak height = pole weight) and a Lorentzian half-width $0.018\,\Omega$, fitted to the G_0W_0 peaks of Golze. Ticks: exact $t_c = 0$ QP and satellite energies (dotted lines). Dashed: the G_0W_0 curve of Ref. [2], digitized from their Fig. 2 and scaled with their exact QP peak (`measure_golze_fig2.py`); its QP peak at $\rho/\Omega = 1.0$ is cut off by their frame.

this model, whose QP pole is within 0.04 and 0.23 of the exact one (-9.59, -8.76 against -9.55, -8.54) while the gGut $B_c = 3$ pole is at -9.93, -9.74 (errors 0.38, 1.21) and the $B_c = 1$ pole at -10.00. The muted poles (dots of Fig. 2) are not satellites: no weight and arbitrary positions, as discussed above.

7 Conclusions

1. **gGut does not reproduce the satellite structure of the model.** For $B_c = 1$, 3, 5, $\rho/\Omega = 0.5$, 1.0 ($V/g = 0.707$, 1.414) and $t_c/g = 0.05$–0.5, the hole spectral function has a single pole with weight 0.977–0.9999, whereas the exact spectrum has satellites near -12 and at -14.45, -15.47 carrying 17.5% and 37.8% of the weight. The main pole is displaced from the exact one (-9.93 vs. -9.55; -9.74 vs. -8.54).

2. The ghosts improve the ground state: $E - E_{\mathrm{ED}} \approx 10^{-7}$ at $B_c = 3$ and 5 (vs. $1.3 \times 10^{-3} V^2 t_c^2$ at $B_c = 1$), ν_c to $\sim 10^{-8}$; $B_c = 5$ returns the $B_c = 3$ energy and spectrum (the extra ghosts

stay asleep).

3. The muted poles (the levels of H_{qp} beyond the three QP levels) have zero weight (R_c lies along the core orbital) and arbitrary energy (free multiplier directions): moving them to the exact satellite energies leaves E and the weights unchanged, so they give no satellite. A solution with satellite weight would have to change R_c, i.e. differ from the energy minimum; none was found.

4. The model uses a core–virtual hopping: a core–occupied hopping leaves $|220\rangle$ exact for every t_c and cannot make R_c determined. The energy functional and its gradients are verified (finite differences $\sim 10^{-10}$), and the limits $V = 0$, $t_c = 0$ are reproduced.

5. The model is the three-level model of Cederbaum *et al.* [1] with $\rho = \sqrt{2}V$; for $\rho/\Omega = 0.5$, 1.0 the exact spectrum has the QP peak and two satellites, G_0W_0 one satellite at too high a binding energy, GW+C satellites at $-n\Omega$, and gGut a single pole (Fig. 4, Tab. 4).

A The quasiparticle Hamiltonian as a matrix

H_{qp} acts on the B_c+2 quasiparticle orbitals of one spin, ordered as the B_c core orbitals $d_{c1}, \ldots, d_{cB_c}$, then d_o, then d_v (the block structure c, o, v of ρ in Sec. 3). It is real and symmetric. Its ingredients, all defined in Sec. 3, are the renormalization vectors $R_c \in \mathbb{R}^{B_c}$, $R_o, R_v \in \mathbb{R}$, the core occupation ν_c, the tensor $\bar{\Theta} \in \mathbb{R}^{B_c \times B_c}$ and the two core vectors

$$a = \bar{\Theta}\, u = \bar{\Theta}\, \rho^{cv} R_v, \qquad b = \bar{\Theta}\, w = \bar{\Theta}\, \rho^{co} R_o \qquad (a, b \in \mathbb{R}^{B_c}). \tag{21}$$

The diagonal blocks are the multipliers of the constraint $\rho^{II} = {}^{I}\Delta$ of Eq. (9), $-\lambda_c \in \mathbb{R}^{B_c \times B_c}$ (symmetric) and $-\lambda_o, -\lambda_v \in \mathbb{R}$; the off-diagonal blocks are $H^{IJ} = \frac{1}{4}\partial E/\partial \rho^{IJ}$ of Sec. 4. In block form, with the blocks of size B_c, 1, 1,

$$H_{qp} = \begin{pmatrix} -\lambda_c & -V R_o\, a & R_v\,(t_c R_c - V\, b) \\ -V R_o\, a^T & -\lambda_o & -V(2-\nu_c)\, R_o R_v \\ R_v\,(t_c R_c - V\, b)^T & -V(2-\nu_c)\, R_o R_v & -\lambda_v \end{pmatrix}. \tag{22}$$

The *co* and *cv* blocks are the column vectors $-V R_o\, a$ and $R_v(t_c R_c - V b)$ ($B_c \times 1$); the *ov* block is a number. This is what `build_Hqp` returns (`gga_cederbaum.py`), and it is the same operator as $H_{\text{off}} - \text{blockdiag}(\lambda_I)$ of Sec. 4.

Spin. H_{qp} and ρ in this appendix are the *one-spin* matrices, of size $B_c + 2$: $\rho_{pq} = \langle d^\dagger_{p\sigma} d_{q\sigma} \rangle$ with σ fixed, and the solutions are paramagnetic, so $\uparrow$ and $\downarrow$ have the same matrix. The quasiparticle Hamiltonian on the full space of $2(B_c + 2)$ spin orbitals, ordered ($\uparrow$ block, $\downarrow$ block), is

$$H^{\text{full}}_{qp} = \begin{pmatrix} H_{qp} & 0 \\ 0 & H_{qp} \end{pmatrix} = \mathbb{1}_2 \otimes H_{qp}, \qquad \rho^{\text{full}} = \mathbb{1}_2 \otimes \rho, \tag{23}$$

and each spin channel fills its N_{occ} lowest levels ($N_{\text{occ}} = 2$ for $B_c = 1$, 3 for $B_c = 3$), i.e. $2N_{\text{occ}}$ electrons in total. The spin sum and the hermitian conjugate that give the factor 4 in Eq. (7) are removed in $H^{IJ} = \frac{1}{4}\partial E/\partial \rho^{IJ}$, so the entries below carry no spin factor.

Explicit form for $B_c = 1$. There are no ghosts, λ_c, a, b, R_c, $\bar{\Theta}$ are numbers and H_{qp} is 3×3:

$$H_{qp}^{B_c=1} = \begin{pmatrix} -\lambda_c & -VR_o\, a & R_v(t_c R_c - V\, b) \\ -VR_o\, a & -\lambda_o & -V(2-\nu_c)R_o R_v \\ R_v(t_c R_c - V\, b) & -V(2-\nu_c)R_o R_v & -\lambda_v \end{pmatrix}, \tag{24}$$

with $a = \bar{\Theta} R_v \rho^{cv}$ and $b = \bar{\Theta} R_o \rho^{co}$.

Diagonalizing it, $H_{qp}\phi_n = \epsilon_n \phi_n$ ($\epsilon_1 \le \epsilon_2 \le \epsilon_3$, $\phi_{n,p}$ the component of ϕ_n on orbital p), and filling the $N_{\mathrm{occ}} = 2$ lowest levels per spin gives $\rho_{pq} = \langle d^\dagger_{p\sigma} d_{q\sigma} \rangle = \sum_{n=1}^{2} \phi_{n,p}\phi_{n,q} \equiv s_{pq}$:

$$\rho^{B_c=1} = \begin{pmatrix} s_{cc} & s_{co} & s_{cv} \\ s_{co} & s_{oo} & s_{ov} \\ s_{cv} & s_{ov} & s_{vv} \end{pmatrix}. \tag{25}$$

The diagonal entries are the new ${}^I\Delta$, and the off-diagonal ones give the a, b of the next H_{qp} through $u = R_v s_{cv}$, $w = R_o s_{co}$: $a = \bar{\Theta} R_v s_{cv}$, $b = \bar{\Theta} R_o s_{co}$.

Explicit form for $B_c = 3$. With $\lambda_c = (\lambda_{ij})$ symmetric, $a = (a_1, a_2, a_3)^T$, $b = (b_1, b_2, b_3)^T$, $R_c = (R_{c1}, R_{c2}, R_{c3})^T$ and $\tau_i = t_c R_{ci} - V b_i$, H_{qp} is 5×5:

$$H_{qp}^{B_c=3} = \begin{pmatrix} -\lambda_{11} & -\lambda_{12} & -\lambda_{13} & -VR_o a_1 & R_v\tau_1 \\ -\lambda_{12} & -\lambda_{22} & -\lambda_{23} & -VR_o a_2 & R_v\tau_2 \\ -\lambda_{13} & -\lambda_{23} & -\lambda_{33} & -VR_o a_3 & R_v\tau_3 \\ -VR_o a_1 & -VR_o a_2 & -VR_o a_3 & -\lambda_o & -V(2-\nu_c)R_o R_v \\ R_v\tau_1 & R_v\tau_2 & R_v\tau_3 & -V(2-\nu_c)R_o R_v & -\lambda_v \end{pmatrix}. \tag{26}$$

Diagonalizing it ($\epsilon_1 \le \cdots \le \epsilon_5$) and filling the $N_{\mathrm{occ}} = 3$ lowest levels per spin gives $\rho_{pq} = \sum_{n=1}^{3} \phi_{n,p}\phi_{n,q} \equiv s_{pq}$:

$$\rho^{B_c=3} = \begin{pmatrix} s_{c_1c_1} & s_{c_1c_2} & s_{c_1c_3} & s_{c_1o} & s_{c_1v} \\ s_{c_1c_2} & s_{c_2c_2} & s_{c_2c_3} & s_{c_2o} & s_{c_2v} \\ s_{c_1c_3} & s_{c_2c_3} & s_{c_3c_3} & s_{c_3o} & s_{c_3v} \\ s_{c_1o} & s_{c_2o} & s_{c_3o} & s_{oo} & s_{ov} \\ s_{c_1v} & s_{c_2v} & s_{c_3v} & s_{ov} & s_{vv} \end{pmatrix}. \tag{27}$$

The upper-left 3×3 block is the new ${}^c\Delta$ and s_{oo}, s_{vv} the new ${}^o\Delta$, ${}^v\Delta$. The off-diagonal entries give the a, b of the next H_{qp}: $u = R_v(s_{c_1v}, s_{c_2v}, s_{c_3v})^T$, $w = R_o(s_{c_1o}, s_{c_2o}, s_{c_3o})^T$, $a = \bar{\Theta} u$, $b = \bar{\Theta} w$.

For $B_c = 5$ the core block is 5×5 and the two border columns have five entries each, same pattern.

Use in the impurity loop. In the impurity loop (`gga_cederbaum.py`) H_{qp} is diagonalized once per macroiteration; there is no inner loop on ρ. The ρ obtained from the occupied eigenvectors is used directly in the next macroiteration: its diagonal blocks ρ^{II} are the ${}^I\Delta$ that enter the embedding problems (before the linear mixing with their output), and its off-diagonal blocks give u, w and hence a, b in the next H_{qp}. ρ is thus self-consistent with H_{qp} only at convergence of the macroiterations. (Repeating the diagonalization until ρ stops changing, before the embedding step, was tested and does not change the converged result for $B_c = 1$ nor make the $B_c = 3$ loop converge.)

In the direct minimization (Sec. 4) no such cycle is needed: $\rho = X(X^TX)^{-1}X^T$ is a variable of the minimization, E already contains the dependence of H_{qp} on ρ through u and w, and the

condition that ρ is built from its own H_{qp} is the stationarity condition $[H_{qp}, \rho] = 0$; the link $\rho^{II} = {}^{I}\Delta(\Phi_I)$ is an equality constraint. At the minimizers found, building H_{qp} from the multipliers and from this ρ, diagonalizing it and filling the N_{occ} lowest levels returns the same ρ to the accuracy of the stationarity ($\|\rho_{\mathrm{filled}} - \rho\| = 10^{-8}$ for $B_c = 1$ and for $B_c = 3$ at $\varepsilon = 10^{-5}$; 2×10^{-5} for $B_c = 3$ at $\varepsilon = 10^{-9}$, where $\|[H_{qp}, \rho]\| = 3 \times 10^{-4}$), and the occupied levels are the lowest ones, with a gap 1–2 to the first empty level ($V/g = 1$, $t_c/g = 0.5$).

References

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